

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA0101

AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I G.Karthik Reddy-192525047 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Karthik Reddy

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	

AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent

		banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.

10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

AI-Based Medical Diagnosis System (Healthcare Domain)

1. Title of AI Application

AI-Assisted Medical Diagnosis System

2. Domain Selection

Healthcare

3. Introduction and Background

Artificial Intelligence (AI) is transforming healthcare by helping doctors diagnose diseases quickly and accurately. AI-assisted medical diagnosis systems analyze patient data, medical images, and laboratory reports to detect diseases such as cancer, diabetes, heart disease, and pneumonia. This improves the quality of healthcare and reduces human errors.

4. Problem Identification

Traditional diagnosis mainly depends on doctors' experience and manual analysis. This process can be slow and may lead to incorrect diagnoses, especially when handling a large number of patients. Delayed diagnosis can affect treatment and patient recovery.

5. Need for AI Solution

AI can process large amounts of medical data in a short time. It helps doctors detect diseases earlier, improves diagnostic accuracy, reduces workload, and supports better treatment decisions. AI acts as a decision-support tool rather than replacing doctors.

6. AI Technologies Used

- Machine Learning (ML)
- Deep Learning (DL)
- Computer Vision
- Natural Language Processing (NLP)
- Artificial Neural Networks (ANN)

7. Working Mechanism

- Patient information, medical images, and lab reports are collected.
- The data is cleaned and processed.
- AI algorithms analyze the data and compare it with previously learned patterns.
- The system predicts possible diseases and provides diagnostic suggestions.
- Doctors verify the results and make the final treatment decision.

8. Data Sources and Processing

The system uses:

- Electronic Health Records (EHR)
- X-rays, CT scans, MRI images
- Blood test reports
- Patient medical history
- Clinical databases

The collected data is cleaned, organized, and processed before being used for AI prediction.

9. Real-World Implementation

AI-assisted medical diagnosis is used by:

- Hospitals and diagnostic centers
- Google DeepMind Health
- IBM Watson Health
- Mayo Clinic
- Apollo Hospitals (India)

These organizations use AI for disease detection, medical imaging, and clinical decision support.

10. Impact and Benefits

- Faster diagnosis
- Higher diagnostic accuracy
- Early disease detection
- Reduced workload for doctors
- Better patient care
- Lower healthcare costs
- Improved treatment planning

11. Challenges and Limitations

- Patient data privacy and security
- High implementation cost
- Need for large, high-quality datasets
- AI bias due to limited training data
- Dependence on internet and computing resources
- AI cannot completely replace human doctors

12. Future Scope and Emerging Trends

- Personalized medicine using AI
- AI-assisted robotic surgery
- Wearable health monitoring devices
- AI-based remote healthcare services

- Faster drug discovery using AI
- More accurate disease prediction models

13. Conclusion

AI-assisted medical diagnosis is one of the most important applications of Artificial Intelligence in healthcare. It improves diagnostic accuracy, saves time, supports doctors in decision-making, and enhances patient care. Although challenges such as privacy and implementation costs remain, AI will continue to play a major role in the future of healthcare.

14. References

1. World Health Organization (WHO) – Artificial Intelligence in Health.
2. Google DeepMind Health.
3. IBM Watson Health.
4. Nature Medicine Journal.
5. IEEE Xplore Digital Library.